DIPLOMA PROGRAMME PRACTICE

# Mathematics

## Applications and interpretation SL

### Set A / Paper 1 / Short response

90 minutes · 80 marks · All questions compulsory

Use a graphic display calculator and a clean Mathematics formula booklet.

Write your reasoning and working in the spaces provided; use additional answer paper if needed. Unless stated otherwise, give exact answers or round numerical results to three significant figures.

Diagrams are illustrative unless coordinates or dimensions are specified.

## QUESTION 1

6 marks

A rectangular glass panel is recorded as 1.24 m by 0.86 m. Each length was rounded to the nearest 0.01 m.

**(a)** State the lower and upper bounds for the longer side, with appropriate inequalities. [2]

**(b)** A frame needs at least 1.060 m² of glass. Determine whether the recorded dimensions guarantee that it will fit this area requirement. [4]

## QUESTION 2

6 marks

In the first week of a trial a library issues 160 passes. In each subsequent week it issues 12 more passes than in the previous week.

**(a)** Find the number issued in week 9. [2]

**(b)** Find the first week by the end of which the total number issued exceeds 2000. [4]

## QUESTION 3

6 marks

A cold-store probe gives a reading of 2.6 V at −4 °C and 5.0 V at 8 °C. Assume that voltage is a linear function of temperature over this interval.

**(a)** Find the voltage model V = mT + c. [3]

**(b)** A reading of 4.3 V is observed. Estimate the temperature and explain why this is interpolation. [3]

## QUESTION 4

6 marks

A rotating sprinkler waters a sector with radius 7.5 m and central angle 128°. Within the sector, water lands uniformly at 3.0 litres per square metre during one cycle.

**(a)** Calculate the length of the curved outer edge. [2]

**(b)** A 190-litre tank is full. Decide whether it can supply one cycle, showing the unrounded calculation used to decide. [4]

## QUESTION 5

6 marks

Seven randomly selected morning journeys took 12, 13, 13, 14, 15, 17 and 28 minutes. Use the median-of-halves convention, excluding the overall median.

**(a)** Find the interquartile range and determine whether 28 is an outlier using the 1.5 × IQR rule. [4]

**(b)** A manager uses the sample mean to advertise a typical journey at any time of day. Give two distinct reasons for caution. [2]

## QUESTION 6

6 marks

A parcel desk processes 300 items: 180 are standard and 120 are express. Of the standard items, 9 arrive late; of the express items, 12 arrive late. One processed item is selected at random.

**(a)** Find the probability that the item arrived late. [2]

**(b)** Given that it arrived late, find the probability that it was express. [2]

**(c)** Determine whether being express and arriving late are independent for these data. [2]

## QUESTION 7

7 marks

For six bus stops the shelter length x (m) and the number y of seats are paired as follows: (2, 4), (3, 5), (4, 7), (5, 8), (6, 11), (7, 13).

**(a)** Use technology to find the least-squares regression line of y on x and the value of Pearson's r. [4]

**(b)** For the stop with x = 5, find the residual (observed minus predicted). Explain its sign. [3]

## QUESTION 8

7 marks

At a loading bay, each crate independently has probability 0.12 of needing a second scan. A batch contains 18 crates.

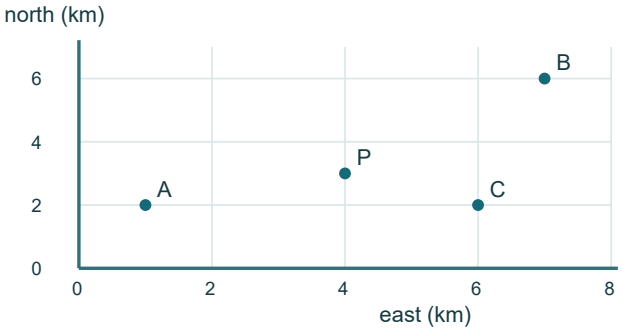
**(a)** Calculate the probability that exactly two crates need a second scan. [3]

**(b)** Extra staff are called when three or more crates need a second scan. Find the probability of calling them, and explain why a change of scanner during a batch could invalidate the model. [4]

## QUESTION 9

7 marks

Two repair depots are located at A(1, 2) and B(7, 6), in kilometres. Travel is modelled by straight-line distance. The boundary contains points equidistant from A and B.




**(a)** Determine the equation of the boundary. [3]

**(b)** A new site C(6, 2) opens. A fault is reported at P(4, 3). Determine whether adding C changes which depot serves P. [4]

## QUESTION 10

7 marks

The amount of water in a tank is $V(t) = 24 + 12t - 1.5t^2$ litres, for $0 \le t \le 6$ minutes.

**(a)** Find the rate of change at $t = 2$, including units. [3]

**(b)** Determine the maximum amount of water during the modelled period and justify that it is a maximum. [4]

## QUESTION 11

8 marks

A co-operative borrows 9600 dollars. Interest is 6% nominal per annum compounded monthly. It makes 36 equal payments at the end of each month; there are no fees and the final balance is zero.

**(a)** Determine the monthly payment to the nearest cent. [3]

**(b)** Find the total interest paid to the nearest dollar, using the unrounded payment. [2]

**(c)** The co-operative considers 48 payments at the same rate instead. Compare the monthly payment and the total interest, explaining the difference. [3]

## QUESTION 12

8 marks

Outflow from a sluice is measured every two minutes. At t = 0, 2, 4, 6 and 8 minutes the rates are 18, 25, 29, 26 and 20 litres per minute respectively.

**(a)** Use the trapezoidal rule with all measurements to estimate the discharged volume. [3]

**(b)** A technician instead uses only the first, middle and last readings. Calculate the revised estimate and its percentage difference relative to the five-reading estimate. [4]

**(c)** Explain why neither estimate establishes the exact volume. [1]